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Webcam Respiration - Getting Started

This article describes study setup, data collection and analysis of Webcam-based respiration detection using iMotions Remote Data Collection. For a detailed validation of the algorithm find the white paper here.

Setting up your study

Webcam-based respiration data is automatically collected for all stimuli in iMotions Remote Data Collection studies. However the feature must be enabled for the account first - contact iMotions if you are uncertain if you have access. Currently, the Webcam Respiration feature is tied to Webcam Eye Tracking. Hence your study must be created as a webcam eye-tracking study to gain access to the respiration data. See article on how to create a Webcam Eye Tracking study - here.

Stimuli Types

The best results are obtained for stimuli that require passive viewing, i.e. no interaction with mouse and keyboard. This typically means video and still images. The system will allow you to use other stimuli types such as websites and surveys, however be aware that data quality will most likely drop for these.

Instructions

The algorithm detects respiratory patterns by measuring subtle movements in the chest area of the participant. Hence unnecessary movement of torso and arms will negatively affect the ability of the algorithm to measure respiratory patterns.

Hence, good, clear instructions to the participants are recommended. Encourage participants to sit still during the experiment. A position where the participant is leaning against the backrest of the chair is recommended, as depicted on this image:

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Data collection

When you have finished designing the experiment in iMotions, you can upload the study to the cloud account for distribution to respondents.
Follow the instructions in this article, under section “Starting Data Collection on my.imotions.com”.

Reviewing the Data

For each participant, you will obtain the following time series data:

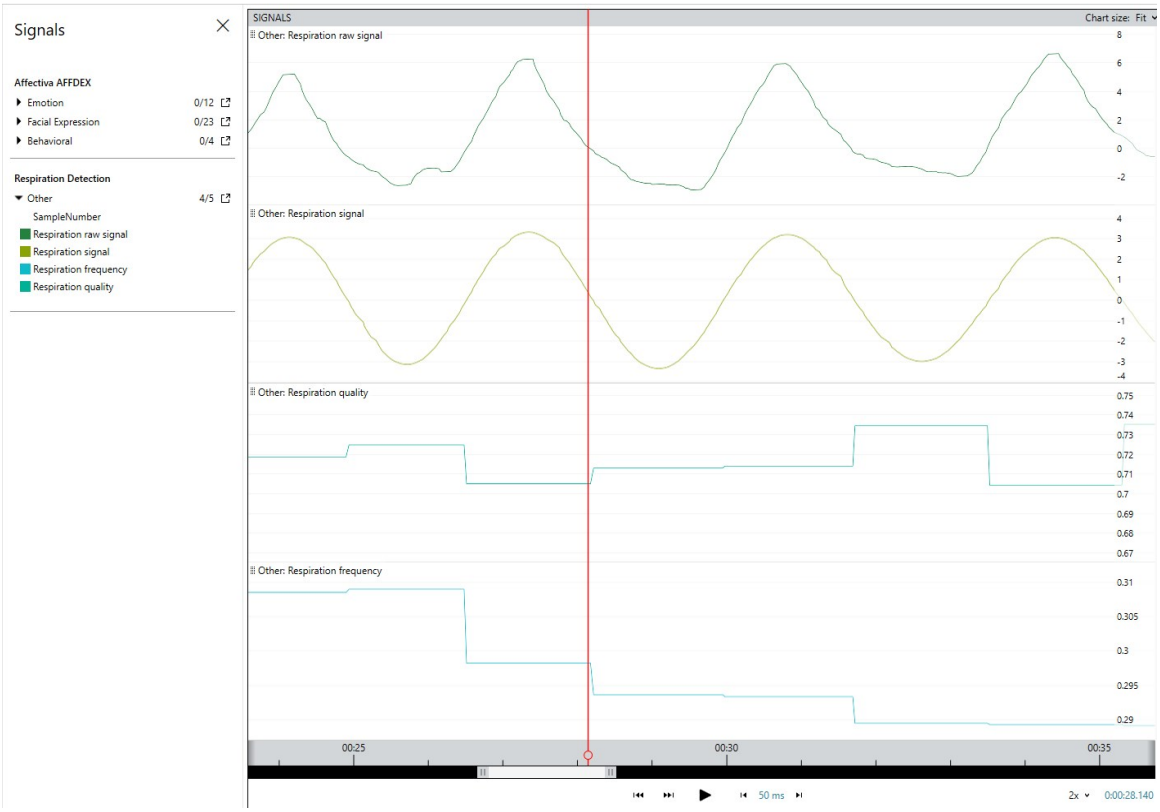
Respiration Raw Signal
The raw detection of chest movements, the scale is unitless.

Respiration Signal
A filtered version of the raw signal, the scale is Unitless. The filtering is using a bandpass of frequencies of expected respiratory rate range of 7.5-30 Breaths Per Minute (0.125 Hz - 0.5 Hz).

Respiration Frequency
Respiratory rate in Hertz. The rate is calculated for each half-cycle. This signal is a preliminary estimation. When applying R-Notebook processing (see below) the Frequency signal is post processed.

Respiration Quality
Quality score for each half respiratory cycle [0-1] Generally a score above 0.5 represents a useful detection of a respiratory cycle.

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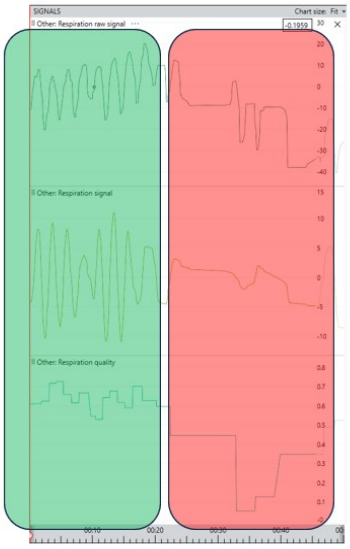


Examples of Good and Bad Respiration Data

Oscillatory pattern corresponding to inhale and exhale.

Visible both in raw signal and filtered signal below

The quality score is generally in the range 0.5-1.0



No evident oscillatory pattern.

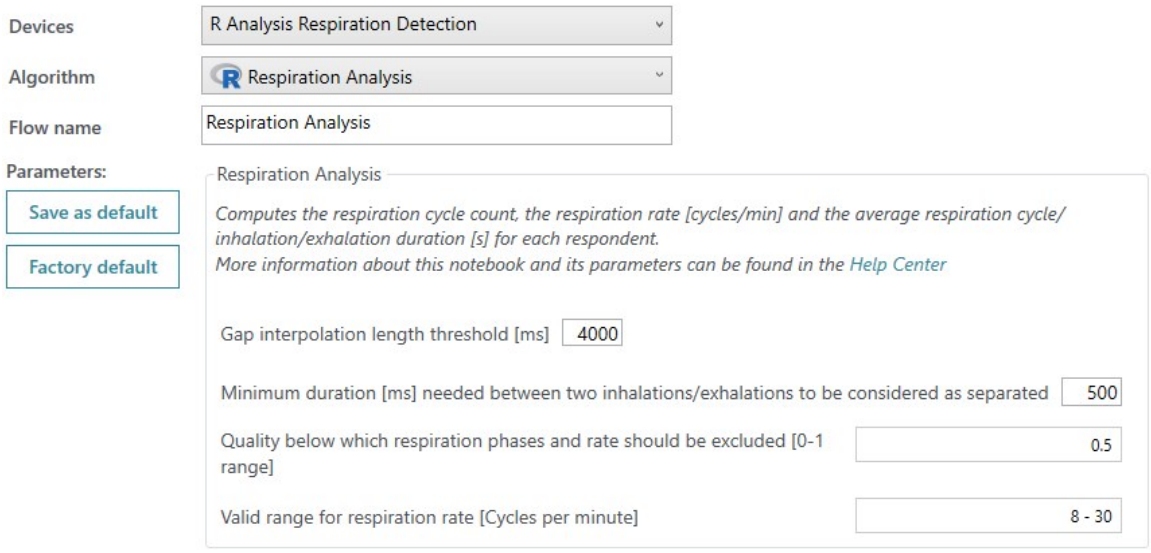
Both in raw signal and consequently in the filtered version of the signal.

The quality score is generally less than 0.5

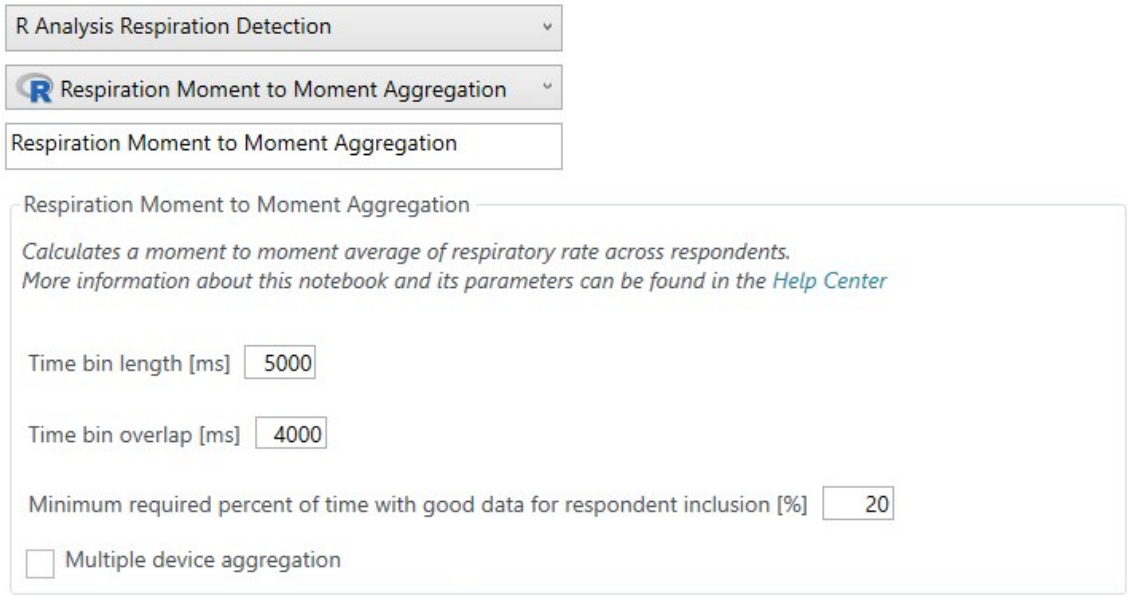
Analyzing the Data

In the dialog where you can choose R Notebooks to process the raw data, these will be the default parameters:

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Default parameters will work for many cases. If you want to improve the detection accuracy you can increase the quality threshold - be aware that the higher the quality threshold, the more data will be excluded from the analysis.
For some stimuli, especially Video stimuli, you can use the aggregation R Notebook to get a moment-to-moment visualisation of the average respiratory rate:



Read more

- R Notebook for respiration detection here.
- R Notebook for moment to moment aggregation here.

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